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SE LAB3

Student Startup Crowdfunding Platform

Step1:

Main purpose:
A platform that allows student founders to create crowdfunding campaigns, receive backer pledges and rewards, and release funds only after faculty committee verification of project milestones.

Main stakeholders/actors:

1. Student Founder
2. Campaign Backer
3. Faculty Committee

5 components:

1. **Campaign Management Component**
2. **Payment & Pledge Component**
3. **Reward Management Component**
4. **Milestone Verification Component**
5. **Fund Release Component**

STEP 2: Architectural Style Analysis

1. Layered Architecture

Structure: The system is organized into horizontal layers such as Presentation, Business, and Data layers.

Advantages:

- Provides clear separation of concerns.
- Easy to understand and maintain.

Disadvantages:

- Can introduce performance overhead.
- Difficult to scale individual components independently.

2. Microservices Architecture

Structure: The system is divided into small, independently deployable services.

Advantages:

- Different functions such as payment, campaign management, and milestone verification can operate independently.
- Individual services can be scaled independently.

Disadvantages:

- More complex to develop and manage.
- Communication between services can introduce network latency and data consistency challenges.

3. Client-Server Architecture

Structure: Multiple clients communicate with a centralized server.

Advantages:

- Centralized control of system data.
- Easier to maintain data consistency.

Disadvantages:

- The central server can become a scalability bottleneck.
- Server failure can affect the availability of the entire system.

STEP 3: Architecture Selection & Component Identification

Selected Architecture

Microservices Architecture is selected for the Student Startup Crowdfunding Platform because the system contains multiple independent functions that can be organized as separate components/services.

Identified Components

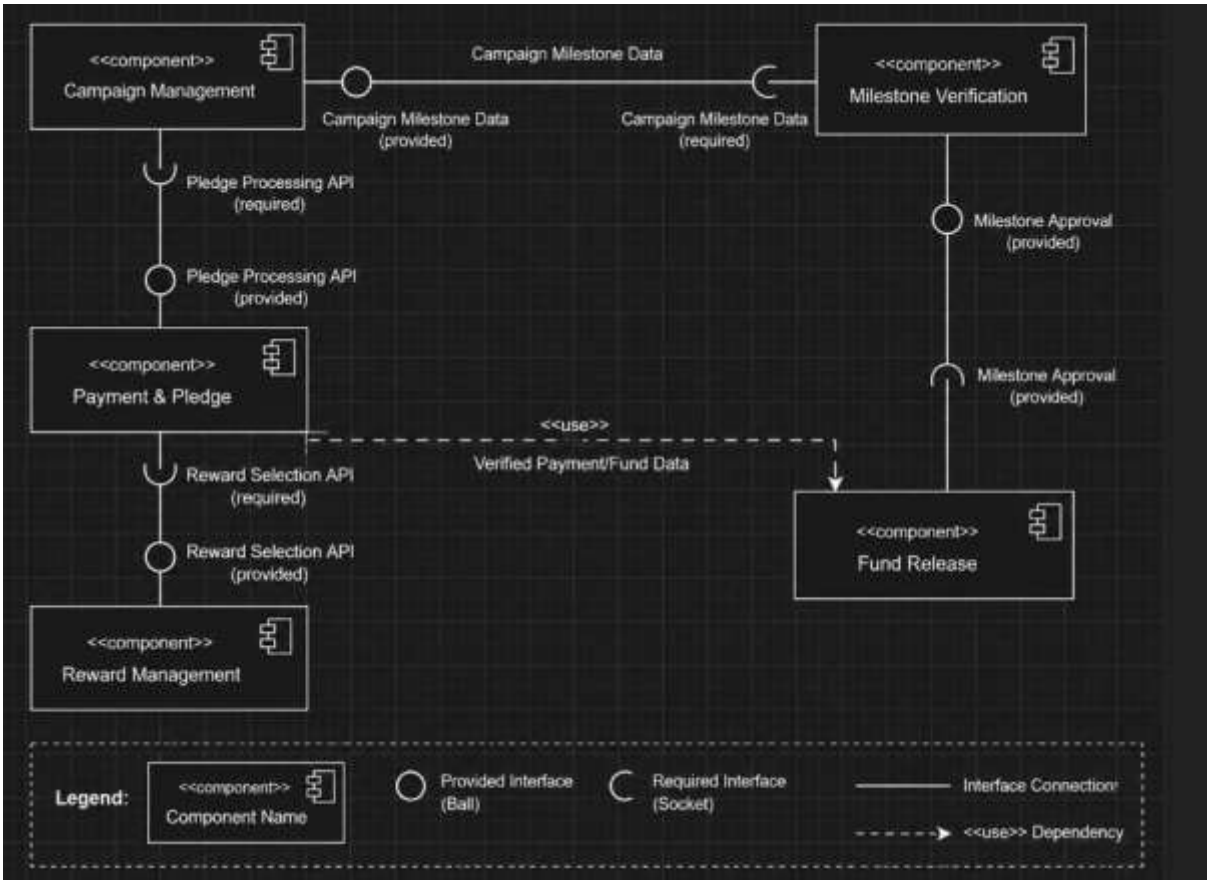
The following five components are identified for the system:

1. **Campaign Management Component**
    - Allows Student Founders to create, update, and manage crowdfunding campaigns.
    - Stores campaign details such as funding targets and campaign information.
2. **Payment & Pledge Component**
    - Handles backer pledges and payment processing.
    - Uses secure and tokenized payment authorization as required by the system.
3. **Reward Management Component**
    - Manages reward tiers selected by Campaign Backers.
    - Associates the selected reward with the corresponding pledge.
4. **Milestone Verification Component**
    - Manages predetermined project milestones.
    - Allows the Faculty Committee to review milestone evidence and approve or reject milestones.
5. **Fund Release Component**
    - Releases campaign funds in tranches only after the required milestone has been verified and approved.
    - Ensures that funds are not released without milestone sign-off.

Main Component Interactions

- Campaign Management communicates with Payment & Pledge to process campaign funding.
- Payment & Pledge communicates with Reward Management to record the reward selected by the backer.
- Milestone Verification communicates with Fund Release to authorize the release of funds after approval.
- Campaign Management provides campaign and funding information required by other components.

STEP4.UML DIAGRAM:



STEP 5: Architecture Justification

Selected Architecture: Microservices Architecture

The Student Startup Crowdfunding Platform uses a **Microservices Architecture** because the system contains multiple independent functions that can be separated into individual components.

The **Campaign Management Component** manages student startup campaigns and their details. The **Payment & Pledge Component** handles backer pledges and secure payment processing. The **Reward Management Component** manages the reward tiers selected by backers. The **Milestone Verification Component** allows the faculty committee to verify project milestones. Finally, the **Fund Release Component** releases funds only after the required milestone has been verified and approved.

The components communicate through defined interfaces such as **Pledge Processing API, Reward Selection API, Campaign Milestone Data, and Milestone Approval**. This

provides clear communication between components while keeping their responsibilities separate.

Microservices architecture is suitable because individual services can be **developed, maintained, and scaled independently**. For example, the Payment & Pledge component can handle a large number of payment requests without requiring the entire system to be scaled. The separation of milestone verification and fund release also helps enforce the rule that funds are released only after faculty approval.

Although microservices can be more complex to manage than a simple layered or client-server architecture, it provides better modularity and flexibility for the crowdfunding platform.